Rehearsal script: SCEBE 2026 talk

Oct 19, 2026 · @Mahmoud Essamuldin Yahia Elrggal

How to use this script

Fifteen slides, one minute each, then five minutes of questions. Rehearse aloud with a timer at least twice and aim to finish at 14 minutes. Each slide section gives the take-away, the words to say, the sentence that moves to the next slide, and the sources on that slide.

The three key messages, in the order the talk builds them:

1. Clinical AI, and sepsis AI in particular, performs well where it was built and fails in the next hospital, because it learns from a narrow, leaky and care-process-driven picture of the patient.
2. This project asks a different question: using only what was known at each hour, can a warning have positive clinical utility, and does that utility hold in a hospital system it has never seen?
3. Healthcare professionals in the loop change what gets built: which inputs are safe, which leak the answer, and which alert a team would act on.

Say "preliminary" every time a result appears (slides 11 to 13): "These are preliminary results on the development split. The full results are in the manuscript we are preparing." Never quote the March numbers; if asked, say the earlier models were retired after a leakage audit.

Every number on the slides carries a bracketed reference; the full list is on the last two backup slides (29 and 30). The numbers below follow the 20 October build; where a number is missing, the author and year are what to know. You do not need to read the references aloud, but know the first author of each key number (listed under each slide below).

Keep the language professional: "as a healthcare professional", never personal labels. Define sepsis, SOFA and utility once (slides 3 and 5) and then use the words without re-explaining.

Slide 1: Title (1 minute)

Take-away: the conference title, and the one question this talk answers.

Say: "Good morning. My name is Mahmoud Elrggal. I am a healthcare professional and a PhD candidate in the School of Computing, Engineering and the Built Environment, supervised by Professor Ahmed Al Dubai and Dr Kia Dashtipour. My PhD is about developing AI models for the early detection and prediction of infectious diseases, with a focus on sepsis. Today I want to answer one question, the one in red: can a sepsis warning stay useful in a hospital it has never seen? I will start with why early detection of infection matters, then what sepsis is and what the medical literature tells us about clinical AI. Then I will show how this project answers the question differently, the preliminary results, and two points where I would value your input."

Move on with: "Let me start with infection in general."

Watch: know the first two sentences by heart. Pause after reading the red question. The title says data-driven: if asked whether that includes text, yes, the text entries clinicians chart, drug names, diagnoses and orders are read by rule from the record; narrative notes such as radiology reports are the planned next phase, with their design fixed. It says mixed-method: the consultants' study is the second method.

Slide 2: Early detection of infection: lives and antibiotics saved (1 minute)

Take-away: infection kills on a huge scale, and acting earlier demonstrably helps; AI can move recognition earlier and diagnosis faster.

Say: "Four numbers, and no sepsis yet. In 2019, 13.7 million deaths were infection-related, and 7.7 million of them were linked to 33 common bacteria: about one death in seven worldwide. In the same year 1.27 million deaths were caused directly by bacteria resistant to antibiotics, and nearly five million were associated with them. Acting earlier helps: in bloodstream infections, rapid molecular diagnosis lowered the odds of death, odds ratio 0.66, and got patients onto effective treatment about five hours sooner. And AI can move detection earlier: in 2021 the World Health Organization recommended computer-aided reading of chest X-rays for tuberculosis screening, after independent testing showed the software matched expert readers. The pathway at the bottom shows where AI helps: it moves recognition earlier, and it makes diagnosis faster."

Move on with: "Sepsis is where that time matters most, so let me explain what it is."

Sources on this slide: GBD 2019 Antimicrobial Resistance Collaborators, Lancet 2022 [1]; AMR Collaborators, Lancet 2022 [2]; Timbrook et al., Clin Infect Dis 2017 [3]; WHO TB screening guidelines 2021 [4]; Qin et al., Lancet Digit Health 2021 [5]; own review, submitted [6].

Watch: if asked about the tuberculosis figure: WHO recommends the software for screening and triage in people aged 15 and over, with a confirmatory test after a positive read. If asked about the 13.7 million: it is infection-related deaths of all kinds; 7.7 million are the deaths associated with the 33 bacteria studied.

Slide 3: Sepsis: infection with organ failure (1 minute)

Take-away: sepsis is infection plus organ failure; it is common, time-critical and hard to recognise early.

Say: "Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection. In plain words, the body's response to an infection starts to damage its own organs. Doctors measure that damage with the SOFA score, which gives zero to four points to each of six organ systems, shown on the left: the lungs, measured by how well oxygen gets into the blood; the circulation, by blood pressure and the drugs needed to support it; the kidneys, by creatinine and urine output; the liver, by bilirubin; clotting, by platelets; and the brain, by the Glasgow Coma Scale. Why is it hard? It is common: sepsis contributes to between one in two and one in three hospital deaths. It is time-critical: each hour of delay to antibiotics raises the odds of death by about four per cent, and in septic shock survival falls by 7.6 per cent per hour. And it is hard to see early: fever, a fast heart rate and fast breathing are common in many ICU patients, and the simple bedside scores detect deterioration less accurately than early-warning scores."

Add, pointing at the two red lines: "The two lines in red are why prediction matters: how lethal, and how fast. The footnote spells out the acronyms: SOFA, Sequential Organ Failure Assessment; SIRS, the 1992 Systemic Inflammatory Response Syndrome criteria; qSOFA, the quick bedside version."

Move on with: "This is exactly the kind of problem people hope AI will solve. So what does the medical literature say about clinical AI?"

Sources on this slide: Singer et al., JAMA 2016 [7]; Vincent et al., Intensive Care Med 1996 [8]; Liu et al., JAMA 2014 [9]; Seymour et al., NEJM 2017 [10]; Kumar et al., Crit Care Med 2006 [11]; Churpek et al., Am J Respir Crit Care Med 2017 [12]; Bone et al., Chest 1992 [13]; Seymour et al., JAMA 2016 [14].

Watch: point at each organ tile as you name it; the engineers will remember the picture.

Slide 4: Clinical AI: strong in development, weak on external validation (1 minute)

Take-away: four recurring weaknesses of clinical AI, each documented in the literature; the rest of the talk is built to address them.

Say: "Before sepsis, the general lesson from medical AI, in four boxes. First, it fails on new data. A systematic review of 86 deep-learning algorithms in radiology found that 81 per cent performed worse when tested on external data. In one silent trial, a model that scored 0.90 in development fell to 0.50, a coin toss, on new patients. And of 81 deep-learning studies compared with clinicians, only nine were prospective and six were tested in real clinical settings. Second, it sees a narrow picture: the most used sepsis benchmark gives models forty hourly variables, leaving out treatments, devices and the patient's history. Third, it learns the care process rather than the patient. A study of 669,000 patients found that whether a laboratory test was ordered predicted survival for 86 per cent of tests, and the time the test was ordered predicted survival better than the result itself for 68 per cent. The record carries the clinicians' decisions, and a model learns them. A pneumonia model learned which hospital took the X-ray rather than the disease. Fourth, it is judged by numbers rather than clinical meaning: data leakage, where the model is given information from after the prediction time or from the answer itself, inflates results in at least seventeen scientific fields, and a prediction is only useful if it triggers an action that helps."

Move on with: "Sepsis AI follows exactly this pattern."

Sources on this slide: Yu et al., Radiol Artif Intell 2022 [15]; Kwong et al., Front Digit Health 2022 [16]; Nagendran et al., BMJ 2020 [17]; Reyna et al., Crit Care Med 2020 [18]; Agniel et al., BMJ 2018 [19]; Zech et al., PLoS Med 2018 [20]; Kapoor and Narayanan, Patterns 2023 [21]; Shah et al., JAMA 2019 [22]; Kelly et al., BMC Med 2019 [23].

Watch: the Agniel finding is the one engineers find most surprising; slow down on it.

Slide 5: Sepsis AI: the same gap, and what AUROC hides (1 minute)

Take-away: the best public competition entry and the most deployed commercial model both failed outside the data they were built on.

Say: "Two concrete cases. On the left, the PhysioNet 2019 Challenge, the largest open competition on this problem: 104 teams, one scoring rule. That rule is called utility. It rewards an alarm raised up to twelve hours before sepsis, most of all around six hours before, charges for every false alarm, and penalises missed sepsis; zero means a system that never alarms. The winning team's official score was 0.36 over three hospitals; on new patients from the two hospitals it trained on it scored 0.43. On a third hospital it had never seen, it scored minus 0.12: worse than having no alarm at all. On the right, the Epic Sepsis Model, deployed in hundreds of American hospitals. The vendor reported an AUROC of 0.76 to 0.83. An independent validation found 0.63, with 67 per cent of sepsis cases missed while the system alerted on 18 per cent of all admissions. Across the field, a systematic review of 28 models found AUROCs from 0.68 to 0.99 with almost no prospective testing, and studies use different definitions of sepsis, so their results cannot be compared. The same three causes recur: information from the future, a narrow set of inputs, and scores that do not tell you whether an alarm would help."

Move on with: "An AUROC cannot show either of those two numbers. A utility score counts every missed patient, every false alarm and the hours gained, in one number. That is the score this project is judged by, and it leads to the question this research asks."

Sources on this slide: Reyna et al., Crit Care Med 2020 [15]; Morrill et al., Crit Care Med 2020 [21]; Wong et al., JAMA Intern Med 2021 [22]; Fleuren et al., Intensive Care Med 2020 [23]; Moor et al., Front Med 2021 [24].

Watch: say "utility" slowly the first time; slides 12, 13 and 15 depend on it. If asked: the winner's overall official score across all three hospitals was 0.36.

Slide 6: The research question (1 minute)

Take-away: this is the research problem; if the audience remembers one slide, it is this one.

Say: "Here is the question, and I will read it as it stands. Using only what a clinician could have known at each hour, can a model warn of sepsis hours before onset with positive clinical utility, and does that utility hold in a hospital system it has never seen? Compare it with what has mostly been asked: how high can the AUROC go on one prepared dataset, with one team's definition of sepsis. This project asks whether the alarm would have helped this patient, at a false-alarm burden clinicians accept, and whether it would still help in the next hospital. Three research questions follow. Predictors: does the whole clinical picture known at that hour, treatments, devices, admission context and history, add signal beyond vital signs and laboratory values? Architecture: once causality is enforced, does organ-structured modelling add anything over simpler models? And transfer and trust: does positive utility survive in another hospital system, and would healthcare professionals act on the alert?"

Move on with: "The last part of that question, whether professionals would act, shapes how the project is built."

Sources on this slide: Fleuren et al. 2020 [23]; Moor et al. 2021 [24].

Watch: slow down on the question. If behind time, give the three research questions as one word each: predictors, architecture, transfer and trust.

Slide 7: Healthcare professionals in the loop (1 minute)

Take-away: the evidence says clinical AI works better with healthcare professionals involved, and two examples show what that changes in this project.

Say: "Four pieces of evidence. Better together: in a study of over 300 clinicians reading skin lesions, AI support improved diagnostic accuracy beyond either the AI or the clinicians alone. Clinicians acting on AI find more disease: in randomised trials of colonoscopy, endoscopists working with an AI detector found pre-cancerous polyps in 37 per cent of patients against 25 per cent without it; the AI points, the clinician decides. Trust is built, not assumed: clinicians interviewed about a deployed sepsis system trusted it through experience, endorsement from colleagues, and a design that respected their judgement. And responsible clinical AI needs clinical input at every stage, partly because, as we saw, health records encode the decisions of the care team. Two examples of what that means here. Antibiotics: most pipelines either feed every antibiotic order to the model, which then predicts sepsis from the treatment that defines it, or drop antibiotics altogether. A professional reading the order sees the difference: cefazolin given at induction of anaesthesia, stopped within 24 hours, with no culture sent, is surgical prophylaxis, not suspicion of sepsis. We keep prophylaxis as a predictor by guideline rule and exclude empirical and culture-directed antibiotics. Suspicion proxies: a stress-dose hydrocortisone order or a chest X-ray request usually means the team already suspects sepsis. We declare these as proxies and report every result with and without them, so the patient's signal can be separated from the team's."

Say the two examples this way: "Example one, antibiotics. Usually they are either fed to the model, which then predicts sepsis from the treatment given for sepsis, or thrown out altogether. A clinician can tell two kinds apart. One dose as the anaesthetic starts, stopped within a day, no culture sent: that is routine surgical prevention, and it tells the model the patient has just had an operation. Antibiotics started because infection is suspected are part of the sepsis definition, so they are never an input. Example two, suspicion proxies. A chest X-ray request or a stress-dose steroid order looks like a strong predictor; mostly it means the team already suspects sepsis. Every result is reported with and without them, so we can tell how much of the signal comes from the patient and how much from the team."

Move on with: "Now the data this is built on."

Sources on this slide: Agniel et al. 2018 [19]; Tschandl et al., Nat Med 2020 [28]; Hassan et al., Gastrointest Endosc 2021 [29]; Henry et al., NPJ Digit Med 2022 [30]; Wiens et al., Nat Med 2019 [31]; Bratzler et al. 2013 [32]; Evans et al., Intensive Care Med 2021 [33].

Watch: this slide is where your clinical background shows; let the examples speak rather than describing yourself.

Slide 8: Chosen databases (1 minute)

Take-away: three open, credentialed intensive-care databases; two raw hospital systems for the transfer test and the challenge set for comparison.

Say: "Three databases, all from PhysioNet, the open repository for physiological research data. MIMIC-IV, version 3.1, is the de-identified record of the intensive care units of one large teaching hospital in Boston, the Beth Israel Deaconess Medical Center: about 94,000 ICU stays, with every charted vital sign,

laboratory result, medication, procedure, microbiology result and diagnosis. eICU is different in kind: 200,859 ICU admissions from 208 hospitals across the United States in 2014 and 2015, recorded through a tele-ICU programme, with very different charting habits from hospital to hospital and every time stored as minutes from ICU admission. These two are my development system and my second, unseen hospital system. The third is the PhysioNet 2019 Challenge set, about 40,000 patients reduced by the organisers to forty hourly variables with a pre-shifted sepsis label; it lets me compare with the published challenge models. Access requires credentialing, training in human-subjects research and a signed data use agreement, and no patient-level data ever leave the analysis environment. To apply exactly the same definition of sepsis to both raw databases, I use a published open-source benchmark, Yet Another ICU Benchmark, which defines the Sepsis-3 label on both. Only the labels come from it; the predictors are our own."

Move on with: "Here is how everything fits together."

Sources on this slide: Reyna et al. 2020 [15]; Johnson et al., Sci Data 2023 [30]; Pollard et al., Sci Data 2018 [31]; Goldberger et al., Circulation 2000 [32]; van de Water et al., ICLR 2024 [33]; Bennett et al., GigaScience 2023 [34].

Watch: if asked why not other hospitals: these are the largest open ICU databases with credentialed access; testing on a live hospital is the future direction on slide 14.

Slide 9: The research pipeline (1 minute)

Take-away: the whole method is one pipeline, read left to right like a model architecture, with a reason under every step; it feeds a qualitative phase and, in future, hospital patients.

Say: "This is the research drawn the way we draw a model. Phase one is quantitative. Raw records from MIMIC-IV and eICU, because two hospital systems are what make a transfer test real. Then harmonisation: every measurement is translated into the same concept, the same unit and the same clock, so the model compares like with like; I will show what that means on slide 12. Then one label: the hour of sepsis onset by the same Sepsis-3 definition in both databases, so results are comparable. Then hourly features: 659 predictors, each used only from the hour it became available to the clinician, so no information from the future reaches the model. Then models: twenty-one of them, compared on identical hours, because architecture is a hypothesis. Then evaluation by what would help patients: utility, false alarms per patient-day, transfer to the other database, and a sealed holdout of twenty per cent of patients that is opened only once. The evaluation feeds phase two, the qualitative study, which is running now and nearly complete: through online interviews and round-table discussions, twenty infectious disease and critical care consultants from different hospitals tell us whether they would trust and act on the alert, what explanation they need, and what alarm burden is acceptable. Phase three is the future direction: a silent test on hospital patients, where the model runs on live ICU data and its alerts are logged but not shown, then compared with what happened, before any clinician-facing pilot."

Move on with: "Inside the models box, the architecture question."

Sources on this slide: Kwong et al. 2022 [13]; Henry et al. 2022 [26]; Seymour et al., JAMA 2016 [35].

Watch: gesture along the boxes left to right, then down to phases two and three.

Slide 10: Models compared (1 minute)

Take-away: OSTA, the organ-structured architecture from this PhD, is tested against the same model without organ structure and against the published alternatives, with decision rules fixed in advance.

Say: "On the left is OSTA, organ-system-aware temporal attention, as it is built. Five stages, left to right. One: every hour, 659 predictors, each as a value, a flag for whether it was measured, and its age, grouped by organ system, the six SOFA organs and eight other groups. Two, the red box: attention over time. Each organ group has its own small transformer that reads that organ's past hours and decides which of them matter for the current hour, and it can only look backwards. The strip below shows it for the kidneys: the weights are high where creatinine rose and where urine output has been falling, near zero in the quiet hours, and the future is masked. These weights are illustrative, drawn to explain the mechanism. Three: cross-organ attention, at the same hour, where the organ summaries inform each other, so the kidney picture can change how the circulation picture is read. Four: a second transformer reads the whole stay. Five: the outputs, a sepsis risk for the next six hours and a risk per organ, which says which organ is driving it. Why it is different, in one breath: a flat model sees 659 numbers with no time, a recurrent model keeps one running summary, OSTA reads each organ's history, lets organs inform each other, and says which organ drives the risk. On the right, what it is compared with, on identical hours, with the rules fixed in advance: the same model without the organ grouping, which decides whether the organ idea stays; the earlier CNN-Transformer; GRU-D and a temporal CNN from the literature; the competition winner's approach; self-supervised pretraining on unlabelled stays; and the two simple baselines."

Move on with: "Now the preliminary results."

Sources on this slide: Vincent et al. 1996 [8]; Morrill et al. 2020 [24]; Vaswani et al., NeurIPS 2017 [39]; this work [40]; Che et al., Sci Rep 2018 [41]; Bai et al., arXiv 2018 [42]; Yèche et al., ICML 2021 [43]; Yèche et al., ICML 2023 [44].

Watch: if asked why not a bigger transformer: on the current sample size the models overfit within one to three epochs and cannot yet be ranked reliably (backup slide 24); if asked about a specific model, backup slides 21 and 22; if asked why so many models, backup slide 20.

Slide 11: Sepsis onset in the ICU: the cohort study (1 minute)

Take-away: the results open with the submitted cohort paper: who develops sepsis, when, and which routine measurements predict it hours earlier; each predictor is weak on its own, which is the case for hourly models that read everything.

Say: "The results start with the cohort study, submitted to Frontiers in Medicine, on the PhysioNet 2019 ICU patients. Descriptive first: 27,101 patients, 8 per cent of whom developed sepsis, with a median time to onset of 100 hours on the Kaplan-Meier curve. Patients who went on to sepsis had reached the ICU later, 81 hours from hospital admission against 48, and stayed longer. The standard scores, APACHE II, SOFA, NEWS and SIRS, were higher in the sepsis group and rose over the stay. Then the regression: a Cox model with the label moved six hours earlier, to find early predictors. SIRS, APACHE II, male sex, heart rate, respiratory rate and potassium were independent early predictors, each with a hazard ratio close to one. The message in the red band: the signal is there in routine measurements hours before

onset, but it is spread across many weak predictors. That is the case for models that read the whole picture, hour by hour, which is what the next slides test."

Move on with: "Before any model, the two databases had to speak the same language."

Sources on this slide: Elrggal et al., submitted, Frontiers in Medicine [45]; Reyna et al., Crit Care Med 2020 [18].

Watch: these are hazard ratios per unit (per point of SIRS, per beat per minute), so do not read 1.02 as "2 per cent" without saying "per beat". If asked why PhysioNet data here: the study trains no model, so leakage does not arise; it describes and it regresses.

Slide 12: Two hospital systems, one language (1 minute)

Take-away: the same patient looks different in each database until every raw entry is translated into one shared concept, unit and clock; doing that recovered most of the missing data.

What "speak the same language" means (know this well): each database records the same clinical fact in its own way. They differ in four things.

- **Where it is stored.** MIMIC-IV keeps vital signs in one large charting table identified by item numbers. eICU spreads them over a periodic vital-sign table, an aperiodic table for cuff pressures, and a nurse-charting table with free-text labels.
- **What it is called.** MIMIC-IV item 223761 is "Temperature Fahrenheit" and item 223762 is "Temperature Celsius". In eICU the same measurement is a column of the vital-sign table, or a nurse-charting row labelled "Temperature (C)" or "Temperature (F)".
- **The unit.** Fahrenheit or Celsius, pounds or kilograms, rates per minute or per hour.
- **The clock.** MIMIC-IV stores real date-and-time stamps, shifted for privacy. eICU stores minutes since ICU admission. MIMIC-IV also records when a value was entered into the system, which can be later than when it was measured.

Harmonisation maps every raw entry to one concept, for example "temperature in degrees Celsius". It is placed on one clock, hours since ICU admission, and becomes usable only from the hour it was available. After that, a temperature of 38.5 means the same thing to the model whichever hospital it came from.

Say: "Making two hospital systems speak the same language. Every measurement differs between the databases in four things, and the table shows them for temperature. Where it is stored: MIMIC-IV keeps every vital sign in one charting table; eICU spreads them over a monitor table, a separate table for cuff pressures, and the nurse-charting table, which is where temperature actually lives. What it is called: MIMIC-IV uses item numbers, 223761 for Fahrenheit and 223762 for Celsius; eICU uses text labels. The unit: both scales appear in both databases, and ionised calcium is in millimoles per litre in one and milligrams per decilitre in the other. And the clock: MIMIC-IV has date-shifted timestamps with a chart time and a store time; eICU counts minutes from ICU admission. Harmonised means: every table where a value lives is read, one concept per channel, one unit converted and range-checked, and one clock, hours from ICU admission, with each value usable only from the hour it became available to the clinician. Published pipelines read the monitor tables; we read where the values actually are, for 659 predictors in both databases. The chart shows what that recovered: eICU temperature went from 5 per

cent of patient-hours to 94, blood pressure from 21 to 99. One more thing about the clock: our own earlier pipeline failed that test, it used values before they were available, and it was rebuilt causally. Every number from here on is causal."

Move on with: "Now the main comparison."

Sources on this slide: Johnson et al., Sci Data 2023 [34]; Pollard et al., Sci Data 2018 [35]; this work, preliminary [40].

Watch: if asked why the values were missing before: they were not missing from the database, they were in a table most pipelines do not read. Knowing how nurses chart is what found them.

Slide 13: Utility on own and unseen hospital systems (1 minute)

Take-away: two panels, one honest story: at home, on honest data, every model including the winner's approach scores above zero and none is shown best; away, on an unseen hospital system, we are about zero, not below silence where the winner was, and not shown above it. The sealed holdout decides.

Say: "The main comparison, in two panels, same score. Left, own hospital system: the published winner on its own training hospitals, 0.36 official, 0.43 on those two hospitals, and this project on the MIMIC-IV test split, 1,422 stays, 71 sepsis events, every model scored on exactly the same hours. Eight of the twenty-one are shown. Grey, the two baselines at 0.08 and 0.10. Dark grey, the winner's approach, path signatures plus boosted trees, rebuilt in our causal pipeline on our predictors, 0.15. Red, this project: the temporal CNN at 0.17, GRU-D at 0.19, the model our prespecified rule chose on validation, OSTA and flat OSTA both at 0.23, and the self-supervised model at 0.29. All twenty-one are above zero, with every confidence interval above zero. Two things that does not show: that any model is better than another, because the differences are within what seventy-one events can resolve, and anything about transfer. Right, unseen hospital system: the winner fell to minus 0.12 on a hospital it had never seen. Our counterpart is the MIMIC-IV models scored on eICU patients, at development level: about zero, minus 0.03 to plus 0.04 across twenty models, every interval including zero. Not below silence, where the winner was; not shown above it. That is the problem this PhD exists for, and the sealed holdout is the test that answers it. Last, the rules we fixed in advance: organ grouping stays only if it beats the same model without it on validation, and it did not, so it is reported as not supported; the rules shortlisted GRU-D, the temporal CNN and flat OSTA for the full cohort."

Move on with: "So what has the project contributed so far, and what comes next?"

Sources on this slide: Reyna et al. 2020 [18]; Morrill et al. 2020 [24]; this work, preliminary [40].

Watch: do not claim OSTA "won": its interval (0.13 to 0.34) overlaps the winner's approach (0.06 to 0.22) and most others. Never say "winner's method" as if it were a replication: it is their signature idea with our boosted trees on our predictors. The right panel is development-level transfer, not the holdout; say "about zero" and "every interval includes zero", never "positive". If asked which model is best: the highest point estimate is the self-supervised model at 0.29, but it was not top on validation, so it is not the rule's choice, and it joins the full cohort as a declared exploratory addition. If asked why the winner is 0.36 here and 0.43 on slide 5: 0.36 is the official score over three hospitals, 0.43 the score on its own two; both are on the slide.

Slide 14: Contribution and next steps (1 minute)

Take-away: the contribution is the pipeline and evaluation that make results believable; the path leads to hospital patients.

Say: "Contribution so far, on the left. A causal, cross-database pipeline: one sepsis definition, two hospital systems, 659 clinically defined predictors. Evaluation by clinical utility: false alarms per day, warning time, transfer, and a sealed holdout. Three findings so far: harmonisation recovers the data; on honest data all twenty-one models stay above zero at home; and small samples overfit, so no model can yet be called best. And three manuscripts submitted this year: a review of AI in antimicrobial-resistance diagnostics, a cohort study of the temporal patterns and predictors of sepsis onset in 27,101 ICU admissions, and a systematic review and meta-analysis of infrared thermography in infectious disease. On the right, next steps. Now: complete the qualitative study with twenty infectious disease and critical care consultants, which is nearly done, and finish the eICU arm of the model comparison; the MIMIC-IV arm is done. Next: the full cohort, about 94,000 plus 200,000 stays, with three runs and leave-one-hospital-out testing across eICU. Then: freeze the protocol, open the sealed holdout once, and set the alarm policy using the qualitative findings. And the future direction: a silent test on hospital patients, then a clinician-facing pilot. The numbers you saw today are preliminary; the full results are in the manuscript in preparation."

Move on with: "Which brings me to two questions for you."

Sources on this slide: own submitted manuscripts [5], [42], [43]; Kwong et al. 2022 [13]; Reyna et al. 2020 [15]; this work [41].

Watch: if the chair shows two minutes, say only the "Then" and "Future" lines and move on.

Slide 15: What this work adds, the take-home (1 minute)

Take-away: one sentence the room can carry out, the six things done differently as pairs, and three honest lines on what is known; then thank you.

Say: "What this work adds, in one sentence: an early sepsis warning built on raw hospital data, scored only on what a clinician could have known, and judged by what an alarm costs and gains rather than by AUROC. Six things done differently from common practice. Challenge-prepared tables: raw MIMIC-IV and eICU, two hospital systems. Each study its own sepsis definition: one Sepsis-3 definition on both. Values carried forward and information from the future: every value only from the hour it was available. Forty to a hundred and twenty-five variables: 659 clinically defined predictors, text entries included. Antibiotics as inputs: label-defining antibiotics excluded, prophylaxis kept by rule. AUROC on one split: utility, false alarms per day, warning time, and a sealed holdout. What is known so far: on honest, harmonised data, all twenty-one models keep a positive utility at home; the project's own organ-grouping idea is not supported at this scale, and is reported as found; and whether a warning travels to an unseen hospital system is the open question, which the consultants' study and the sealed holdout answer next. Thank you."

Closing: stop after "thank you". Do not summarise; the chair opens questions.

Sources on this slide: this work, preliminary [40]; the six choices draw on Reyna [18], Moor [27], Kapoor [21], Bratzler [32] and Shah [22] (backup slide 18).

Watch: repeat each question back before answering it, and ask a colleague to note who asked what.

If the chair or the panel asks what input you want (backup slide 26 holds the two questions), your own answers:

Q1, which score should choose: "Utility, at a fixed alarm budget; AUROC is reported but never decides. AUROC scores ranking across every threshold, including ones no hospital would use, and with only 0.65 per cent of hours positive it is dominated by easy negatives (Saito and Rehmsmeier, PLoS One 2015). Utility, like decision-curve analysis (Vickers and Elkin, Med Decis Making 2006), weighs an early alarm against false alarms at the threshold you would actually use. Its weights were set by the challenge organisers, not by clinicians, and that is exactly what the consultant interviews test: whether the false-alarm cost matches what clinicians accept." This is already the rule: decisions use validation utility and AUPRC, never AUROC.

Q2, one label across two systems: "Keep each database's best definition as the primary analysis, as the benchmark does, and harmonise both to the antibiotics-only rule as a sensitivity analysis. If transfer improves under the harmonised label, part of the gap is the label rather than the model. Different sepsis definitions select very different patients even within one database (Johnson et al., Crit Care Med 2018). Treating them as two separate targets would give up the transfer question, which is the point of the PhD."

Backup slides: when to use each

Slide	Use it when someone asks
16 Glossary	what Sepsis-3, SOFA, suspected infection, onset, qSOFA, NEWS2, AKI stage, utility or "causal" mean
17 Utility score	how the challenge score works hour by hour; why 0.22 is not "22 per cent accuracy"
18 Six choices	why this project differs from common practice, row by row
19 The 659 predictors	which predictors exist and what is never used
20 How the models evolved	why there are so many models, why the earlier ones were built, and what led to each new one
21 Every model (1 of 2)	about a specific baseline, the winner's approach, the earlier models or the sequence models: what each does, how it differs, its utility
22 Every model (2 of 2)	about the channel encoders, self-supervised pretraining, the hazard head, the ensemble or the

Slide	Use it when someone asks
	retired models
23 Cohorts and holdout	sample sizes, the sealed 20 per cent, the unlabelled pretraining corpus
24 Overfitting	why the models cannot yet be ranked (fast overfitting, a validation split too small) and why scale is the next step
25 Leakage audit and the causal baseline	how we know the earlier pipeline leaked; why the numbers are lower than in March; the causal PhysioNet baseline (0.22)
26 Questions for discussion	the chair or the panel asks what input you want: which score decides; one label across two systems
27 Publications	dates and status of the three submissions
28 Training and the March plan	the RD6 panel: courses and what became of the March 2026 plan
29-30 References	the full list of the 48 numbered references

For the RD6 review, present the 15 talk slides plus slides 20, 27 and 28.

Backup slide 25: Leakage audit and the causal baseline (removed from the talk; use if asked)

Take-away: we audited our own earlier pipeline, found leakage, retired those models, and the causal numbers are the ones we stand behind.

Say: "If asked: we audited our own earlier pipeline, and I want to be direct about it. When we audited our own earlier pipeline we found three kinds of leakage. Hours after sepsis onset were scored as correct early warnings, so recognising sepsis that had already started counted as predicting it. Values were used from when they were measured rather than when they became available to the clinician. And treatments that are part of the sepsis definition were among the inputs. Those models were retired and are not reported. Rebuilt causally, OSTA on the PhysioNet 2019 test set, as an ensemble of three runs, has a patient-level AUROC of 0.81, an hourly AUROC of 0.80, and a utility of 0.22, or 0.30 at the conformal threshold. How to read that: only 1.6 per cent of hours are positive, so ranking scores look modest even for a useful model, and a utility of 0.22 means the alarms achieved about a fifth of the improvement from never alarming to perfect alarms, under the challenge's scoring rules; it is a score, not a measured clinical benefit. This honest baseline is where every later number starts, and the literature shows this kind of leakage is common across science."

Move on with: (not in the talk; return to the slide you were on)

Sources on this slide: Reyna et al. 2020 [15]; Kapoor and Narayanan 2023 [18]; this work [41].

Watch: do not apologise; finding the leak is the point. If asked for the earlier numbers: they came from models that did not hold up and are not reported.

How the models evolved, in three stages (backup slide 20)

Use this when someone asks "why so many models?" or "why did you build OSTA in the first place?", and at the RD6 panel. The story in one line: each stage was triggered by a finding from the one before.

Stage 1, earlier work on PhysioNet 2019. "I started with a CNN-Transformer: convolution picks up short local trends, attention links them across the stay. It worked numerically but treated every variable alike and could not explain itself. So I proposed OSTA: clinicians assess sepsis organ by organ, as the SOFA score does, so OSTA groups the predictors by organ, reads each organ over time, lets the organs inform each other, and gives a risk per organ a clinician can read. It had its own loss, with focal loss for the rare positive hours, extra weight on the hours near onset, and two organ terms. Then I audited my own pipeline and found information from the future in it. Both models were withdrawn, and OSTA was rebuilt causally. That honest baseline, a utility of 0.22, is modest, and that was the first lesson: test causality before anything else."

Stage 2, V10 on MIMIC-IV and eICU. "OSTA learns how much each organ matters only from the sepsis labels. So I asked whether independent evidence could help: small 'observer' networks, one per organ, trained on a separate group of patients, a little like specialty consults. I tested four ways of using them: as extra inputs, added at the output (output fusion), steering OSTA's cross-organ attention through learned gates, and the same steering without gates as a control. Two controls ran alongside: raw values only, to see whether feature engineering adds anything, and extra labelled patients, to see whether any observer gain is just more data. I also restored the original OSTA loss for fidelity. None of them beat OSTA; output fusion was the best of the family and was kept. The training records showed why: every model peaked within one to three epochs. The sample, not the architecture, was the limit."

Stage 3, V11, the prespecified comparison. "So the next stage did three things. It tested the organ idea directly, with flat OSTA, the same model without organ grouping. It brought in what the literature uses for irregular ICU data: GRU-D, a temporal CNN, and the competition winner's own method. And it tried to use scarce labels better: self-supervised pretraining on unlabelled stays, per-channel encoders, attention to fresh measurements only, a hazard head, and a loss weighted by the utility score itself. Every rule for choosing between them was written down before any result."

If asked "was OSTA a dead end?" "No. OSTA is still in the comparison and it scores like the others. What the evidence says so far is that the organ grouping, as built, does not add utility at 10,000 stays. The full cohort tests it again with nine times more sepsis cases, and the per-organ risk may still matter for explanation, which the consultant interviews will tell us."

The models, one by one, and how they differ

Use this when someone asks about a particular model. Slide 10 is the one-minute version; backup slides 21 and 22 show every model with its utility. Every number here is the normalised utility on the MIMIC-IV development test split, one run, preliminary. All 95 per cent intervals are above zero and they overlap, so never present the order as a ranking.

The one idea that ties them together: every model sees the same 659 predictors on the same hours, and each one changes one thing about how time, organs or missing data are handled. That is what makes the comparison a test of ideas rather than a contest.

Baselines: the floor every model must beat

Logistic regression (0.08). One weighted sum of the predictors at the current hour, turned into a probability. It has no sense of time beyond what the predictors themselves carry (changes from the previous value, ages of measurements). Say: "It is the floor. If a complex model cannot beat it, the complexity is not earning its place."

Boosted trees (0.10) [Friedman 2001]. Hundreds of small decision trees, each one correcting the errors of the ones before. It captures thresholds and interactions, such as low blood pressure mattering more when lactate is high, but it still looks at one hour at a time. In tabular medical data it is often the strongest simple model.

The published benchmark

Signatures plus boosted trees, the winner's approach (0.15) [Morrill 2020]. The PhysioNet 2019 winner turned the recent trajectory of each vital sign into a fixed set of numbers called path signatures, which summarise the shape of a curve, and fed them to boosted trees. Difference from ours: time enters through hand-built summaries, not through a model that learns what to look at. It is their approach, not a replication of their full pipeline: their signature idea, on our 659 predictors, with the same boosted trees as our baseline. Rebuilt causally it stays positive on MIMIC-IV, which is itself a finding.

The earlier models of this PhD

OSTA, organ-system-aware temporal attention (0.23). The architecture proposed in this PhD. Predictors are grouped by organ system, matching the six SOFA organs plus a group for infection, treatment and context. Each group is read over time with attention, which lets the model choose which past hours matter, then the organ groups attend to each other, and the output is a sepsis risk plus a risk per organ. Why it was proposed: it mirrors how an ICU team reasons, organ by organ and then how organs affect each other, and the per-organ risk could tell a clinician where the problem is.

Output fusion (0.25). OSTA plus a set of small "observer" networks, one per organ group, trained on a separate group of patients. Each observer gives its own sepsis risk from its organ's measurements, and these second opinions are added to OSTA's output through a learned weight. Difference from OSTA: independent per-organ second opinions fused at the end, a little like a consult from each specialty.

CNN-Transformer (0.24) [Vaswani 2017]. The second earlier model of this PhD. A convolution first picks up short local patterns over 1 to 12 hours, such as a rising heart rate, then a transformer attends across the whole stay. Difference from OSTA: no organ grouping at all.

New supervised models

Flat OSTA (0.23). OSTA with every predictor in one single group. It changes only one thing, the organ grouping, so it is the decisive test of the organ idea. Result: on validation it beat OSTA (0.28 against 0.12) and on test they tie, so the rule fixed in advance says organ structure is not supported at this sample size.

GRU-D (0.19) [Che 2018]. A recurrent network, which reads the stay hour by hour and keeps a memory. Its special feature is learned decay: as a measurement gets older, the model trusts it less and pulls it towards a normal value. Difference: it models missing and stale data explicitly, which fits ICU records where many values are measured only a few times a day. The validation rule chose it.

TCN, temporal convolutional network (0.17) [Bai 2018]. Stacked convolutions that only look backwards, each layer reaching further into the past. Difference: no attention, so it cannot choose which hours matter; it sees a fixed window. Also chosen by the validation rule.

Channel encoders and self-supervised pretraining

CNN to OSTA (0.28). A small convolution for each signal separately, reading its value, whether it was measured, and how old it is, before OSTA. Difference: each signal learns its own pattern first, instead of every signal being treated alike.

Event OSTA (0.21) [Horn 2020]. Each hour attends only to fresh measurements of the fast vital signs in the last 24 hours. Difference: it ignores carried-forward values and uses only what was actually measured, closer to how a nurse reads a chart.

Self-supervised CNN to OSTA (0.29) [Yèche 2021]. The same encoder, but first trained on up to 75,000 unlabelled ICU stays with no sepsis label: forecasting the next value, whether a signal will be measured, and the organ scores 6, 12 and 24 hours ahead. Only then is it fine-tuned for sepsis. Difference: it learns physiology before it sees a single sepsis case, which matters when sepsis cases are few. It had the highest test estimate but was not chosen on validation, so it joins the full cohort as a declared exploratory addition.

Self-supervised CNN, flat head (0.21). The same pretrained encoder without organ grouping. It is the pair that separates the effect of pretraining from the effect of organ grouping.

Pooled encoder twins (0.23 and 0.20). The same models with the encoder pretrained on both databases together. They test whether learning from two hospital systems helps adaptation.

Hazard head (0.18) [Gensheimer 2019]. Instead of yes or no for the next 6 hours, it predicts a survival curve: the risk of onset within 1, 3, 6, 12 and 24 hours. Difference: it uses when onset happens, not only whether, and could give a clinician a time scale.

Mean ensemble (0.28). The plain average of OSTA, the CNN-Transformer and the self-supervised model, with no learned weights. Averaging often smooths out the errors of single models.

Retired in earlier versions. Two other ways of using the same observer networks: feeding their outputs into OSTA as extra inputs (observer features), and letting them steer OSTA's cross-organ attention (gated and ungated attention). They did not differ from OSTA on the earlier runs, so only output fusion was kept.

If asked "so which is best?" "At this sample size, none can be declared best: every model is above zero and the intervals overlap. The rules fixed in advance send GRU-D, the temporal CNN and flat OSTA, with the baselines, to the full cohort, which has about nine times more sepsis cases. That run will narrow the uncertainty, though it may still leave several models level; the sealed holdout then gives the unbiased estimate."

Likely questions and short answers

Why is your AUROC only about 0.8 when papers report 0.95? On MIMIC-IV the hourly AUROC is 0.72 to 0.84 across models. We score every hour, causally, with 0.65 per cent of hours positive and nothing from the future. Many higher figures come from patient-level classification, balanced cohorts or leaked labels (slides 4, 5 and 12).

What does "speak the same language" mean in practice? Every raw entry is mapped to one concept, unit and clock, and used only from the hour it was available. Temperature on slide 11 is the example; the same was done for 76 channels plus drugs, devices, fluids and diagnoses.

How do you know the earlier models leaked? A line-by-line audit found post-onset hours labelled positive, values used before they were available, and label-defining treatments among the inputs. Rebuilt causally, the same architecture scored lower. The rebuild changed other parts of the pipeline too, so the drop is not a pure measure of the leak, but its direction is what removing one looks like.

You call post-onset labels a leak, but your loss trains on hours after onset. Is that not the same thing? No. In the old pipeline, hours after onset were scored as correct early warnings, so the model got credit for recognising sepsis that had already started. Now no hour after onset is ever scored as a warning. The training loss adds only the onset hour and the three hours after it as extra, weighted examples, which mirrors how the challenge's own utility rewards an alarm up to three hours after onset. The inputs at every hour are still only what was known at that hour.

Why MIMIC-IV and eICU rather than PhysioNet only? PhysioNet 2019 is a prepared table with the organisers' label. The raw databases come from different hospitals and charting systems and let one sepsis definition be applied to both, which makes a transfer test real.

What is utility, and why not sensitivity and specificity? It rewards an alarm up to twelve hours before onset, charges for each false-alarm hour and penalises missed sepsis, combining timing and alarm burden (backup slide 17).

Are these utilities real on such a small sample? That they are above zero, yes: all 21 confidence intervals exclude zero (patient-level bootstrap). Which model is best, no: the intervals overlap and validation and test rankings disagree. The full cohort, with about nine times more sepsis cases and three seeds, and then the sealed holdout, will decide.

Is OSTA better than the winner's approach, 0.23 against 0.15? "Not shown. The intervals overlap, and overlapping intervals do not settle it either way; the right test is a paired comparison on the same patients, which we have the per-stay utilities to run and have not run yet. Until then the honest answer is that no difference has been demonstrated."

Why keep prophylactic antibiotics? Surgical prophylaxis marks a recent operation, which changes baseline risk, and does not define the label. It is identified by agent, timing, duration and the absence of cultures or escalation, following the surgical prophylaxis guidelines.

How does involving healthcare professionals improve the model? It decides which inputs are safe and which leak the answer (slide 7), and the evidence shows clinical AI performs and is trusted better when clinicians engage with it (Tschandl 2020, Adams 2022, Henry 2022).

What is the qualitative study? Online interviews and round-table discussions with 20 infectious disease and critical care consultants from different hospitals, on whether they would trust and act on the alert, the explanation they need and the alarm burden they accept. It is nearly complete and will set the alarm policy before the holdout is opened.

Is the organ structure worth it? Not at this sample size. The rule fixed in advance kept it only if OSTA beat the same model without organ grouping on validation; it did not (0.12 against 0.28), and on test the two tie at 0.23. We report that as found. Flat OSTA goes to the full cohort, where the question can be asked again with enough sepsis cases.

Don't you have the same problem as the winner? Its 0.43 fell to −0.12 on an unseen hospital, and your numbers are internal too. "Yes, and I do not claim otherwise. Our MIMIC-IV results are the counterpart of the winner's 0.43: an internal split, and in fact the same hospital system, because PhysioNet's hospital A is Beth Israel Deaconess, where MIMIC-IV comes from. The counterpart of the minus 0.12 is our sealed cross-database holdout, which has not been opened. The early signal, scoring the MIMIC-IV models on eICU development patients, is around zero: no better than silence, though not below it. That is the problem this PhD is built to study, with harmonisation, one label and leave-hospital-out testing. It is not a problem it has solved yet." If asked why not test on PhysioNet: "Half of PhysioNet is the same hospital as MIMIC-IV, so it would not be an external test; eICU, with 208 hospitals, is."

And on eICU, the second hospital system? The MIMIC-IV models scored on eICU development patients sit around zero (utility -0.03 to 0.04, intervals including zero). That is the transfer problem this PhD exists to study, not a surprise: eICU has a weaker, antibiotic-only label and charts differently. The models trained on eICU itself are still running; the full cohort, leave-one-hospital-out testing and label harmonisation address it. Answer only if asked; it is not on the slides.

Which benchmark defines your sepsis label? "Yet Another ICU Benchmark, van de Water and colleagues, 2024, built on the ricu package. Sepsis-3: suspected infection paired with a rise of two or more in SOFA. We run its code pinned to one version; one defect in it, a crash on empty batches, was patched and the patch recorded with its checksum."

The cohort study uses PhysioNet data; is that not the same leaky data? "It is the challenge data, but that study trains no prediction model: it describes who develops sepsis and fits a Cox model with the label moved six hours earlier. Leakage is a failure of prediction pipelines, not of descriptive statistics. One caveat I volunteer: hospital A of that data is the MIMIC-IV hospital, so the two studies share patients."

Your rule did not pick the self-supervised model, so why is it in the full cohort? Because it had the highest test estimate, and that is exactly why it is declared as a deviation, written down before the full-cohort run. It goes in with its twin without organ grouping, both labelled exploratory. The main conclusions come only from the models the rule chose, and the sealed holdout gives the unbiased answer. Say it plainly: "We broke our own rule once, on the record, and we report it as such."

How does the model decide when to alarm? Two thresholds, both set on a separate calibration group of patients and never on test. For the utility score, the threshold that maximises utility on that group, with an alarm every hour the risk is above it. For the clinical metrics, the threshold that catches the most

sepsis while keeping false alarms to at most half an alarm per patient per day, with a six-hour pause after each alarm. The consultants' views will inform which policy a hospital would use.

Is the protocol written down before the results? Yes. The full protocol is in the project repository, checked line by line against the code and timestamped by version control before the full-cohort run. Anything that departs from it is recorded as a declared deviation: one so far, the self-supervised pair. Four earlier decisions taken after interim results, such as a calibration bug fix and moving to three seeds, are disclosed with their reasons.

Have you looked at your test data? "The development test split, yes, many times while building the pipeline, and that is recorded. That is exactly why there is a second, sealed holdout of twenty per cent of patients, chosen by a fixed hash of the patient identifier before any model existed. It has never been opened and will be opened once, after the protocol is frozen."

What about fairness across patient groups? "Race, ethnicity and insurance are never predictors. A subgroup fairness analysis is planned before the holdout is opened; it is not done yet."

How would the hospital test work? A silent run first: the model scores live ICU data, alerts are logged and not shown, and compared with outcomes. It needs ethics approval, a data-governance agreement and a clinical partner.

If you do not know: "I have not looked at that yet; it is a useful point and I will take it away."

Facts to have exactly right (checked against the code)

These were checked line by line against the code that produced the results. If a question touches one, use these numbers and words.

Topic	Exact fact
Predictors	659: the 655 shared set plus two prophylaxis features and their masks. Some are recorded in MIMIC-IV only (prophylaxis, earlier-admission history) and are missing by design in eICU
When a value can be used	From the hour it became available (MIMIC-IV: the later of charted and stored time). Most values are carried for 24 h with their age; GCS, weight and height do not expire
Models compared	21 on MIMIC-IV: 16 architectures, 4 loss variants and a mean ensemble
Splits	20 % of patients sealed by a fixed hash; of the rest, 20 % test, then 60 % training, 20 % observer-fit, 10 % validation, 10 % calibration; always by patient
Decisions	Made on validation only, by the average rank of validation utility and AUPRC

Topic	Exact fact
Alarm thresholds	Utility threshold for the utility score; a budget of at most 0.5 false alarms per patient-day, with a 6 h pause after each alarm, for the clinical metrics
Confidence intervals	Patient-level bootstrap (200 resamples) for utility, AUPRC and Brier score; AUROC is a point estimate
Training positives	Hours up to 6 h before onset, plus the onset hour and the 3 hours after it as weighted extras; no hour after onset is scored as a warning
Pretraining	Up to 75,000 unlabelled stays per database, no labels of any kind
Full cohort	About 94,000 plus 200,000 stays, three seeds, the rule-chosen models plus the declared exploratory pair
Still to decide before the holdout	Final model list and loss; pretraining data at full scale and at the final test; eICU leave-hospital-out details; the fairness analysis

If asked something more detailed than this table, say: "It is specified in the protocol; I am happy to send the section."

Timing plan and rehearsal checklist

Clock	You should be on	Section
0:00	slide 1	title
1:00	slide 2	context: infection, sepsis, literature
5:00	slide 6	question and approach
10:00	slide 11	preliminary results
13:00	slide 14	contribution and next steps
14:00	slide 15	where I would like your input
15:00	finished	questions begin

Chair signals: at 5 minutes left you should be on slide 11; at 2 minutes left on slide 14; at time up, stop within 30 seconds. If late at the 5-minute signal, give slide 12 in two sentences. If late at the 2-minute signal, go straight to slide 15. Never speed up; cut instead.

Before the session: slides on a USB stick as .pptx and .pdf, uploaded at least 10 minutes early; check the charts on slides 11 and 13 and the figure on slide 9 on the room computer.

- ☐ Full run aloud with a timer, twice, finishing at 14 minutes
- ☐ One run in front of a colleague who writes down the questions they would ask
- ☐ The red question on slide 1 and the question on slide 6 known by heart
- ☐ The "same language" explanation for slide 11 practised in your own words
- ☐ The word preliminary said on slides 11, 12 and 13
- ☐ Second supervisor (Dr Kia Dashtipour) shown on slide 1
- ☐ Supervisor's agreement to show the development-split results